HakaseAI

PRODUCT DOCUMENTATION

The operating system for drug development.

A self-improving engine that replaces the patchwork of point tools across discovery, trials, commercial, care and regulatory — where every prediction ships with its falsifier.

BIODIGITAL TWIN · CLINICAL OPERATIONS · COMMERCIAL INTELLIGENCE · CARE & RWE · REGULATORY SUBMISSIONS

Hakase Platform · One engine. Five modules.
Predict-then-confirm, end to end.

Version 1.0 · June 2026
For client & partner distribution

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01 Platform overview

What Hakase is, in one page

Hakase is a unified intelligence platform for the life-sciences industry — an operating system that runs the full arc of a drug program, from the first in-silico hypothesis to a launched therapy and the evidence that supports it at the bedside.

Modern drug development is run by capable teams using disconnected software. Discovery, clinical operations, commercial, care delivery and regulatory affairs each maintain their own stacks, their own data definitions, and their own idea of what "done" means. Evidence loses its provenance at every hand-off. The result is slower programs, weaker submissions, and AI models that quietly drift out of step with reality.

Hakase is built around a different default: **one engine, one evidence ladder, and one set of gates.** A prediction made in discovery is the same object a regulator reads in the dossier — it never loses its falsifier along the way. Every output the platform produces is paired with the experiment that would prove it wrong, and when that result lands, the engine updates.

40%	$2.1M	85%	92%
Faster trial cycles	**Annual cost savings**	**Patient retention**	**Cross-assay accuracy**
Median across sponsors	Per Phase 2/3 program	Avg. across care deployments	vs. held-out CROs

- **Predict-then-confirm**

  Every output ships with its falsifier and a confirmation slot. Wet-lab and clinical results close the loop automatically.

- **Self-improving engine**

  Orchestration, gates, governance and signed audit live in a shared control plane, so every module inherits the same discipline.

- **Outcomes, not seats**

  Drug-Acceleration-as-a-Service: pay per deliverable, per stage, per trial. Tenant-isolated by default; the engine never trains on your data.

02 The problem we solve

Why tool sprawl quietly taxes every program

Bringing a new therapy to patients is one of the most expensive, slow and failure-prone endeavours in any industry. A large share of that cost is not the science itself — it is the friction between the stages of the science.

Evidence loses provenance at every hand-off

A promising signal in discovery is re-entered, re-interpreted and re-justified as it moves into pre-clinical work, into a trial protocol, into a payer dossier, and finally into a regulatory submission. Each translation is manual, lossy, and hard to audit. By the time a claim reaches a regulator, the chain back to the original experiment is often broken.

"Score and serve" AI rarely learns whether it was right

The default mode of AI in pharma is to rank a list of options and stop there. A team picks one, runs the experiment, and the model is seldom told what actually happened. Without that feedback the model cannot improve, and confidence in it erodes precisely when programs need it most.

Five stacks that do not talk to each other

Discovery, clinical, commercial, care and regulatory each run their own point tools. Definitions diverge, data is duplicated, and no single source of truth survives the full journey from molecule to market. The cost shows up as longer timelines, weaker submissions, and decisions defended again and again that the evidence already justified.

> **THE COST OF FRICTION**
>
> The bottleneck in drug development is increasingly the connective tissue between stages — not any single experiment. Hakase is built to remove that tax.

03 The Hakase approach

One engine, one evidence ladder, one set of gates

Hakase replaces five disconnected stacks with a single self-improving engine. Each module composes its own agents, gates and evidence packs from the same foundation, so a result produced in one stage is directly usable in the next — with its provenance intact.

Falsifiable by default

If a prediction cannot be tested, it does not ship. Every claim carries a falsifier and a confirmation slot, so the platform always knows what would prove it wrong.

Confirmations are ground truth

Wet-lab, CRO and clinical readouts are treated as ground truth and feed back into the engine. Customer data is never used as training fuel.

Compose, don't fork

One engine, many modules. A discovery prediction is the same object a regulator reads in the dossier — no re-keying, no lost lineage.

Bench-aware AI

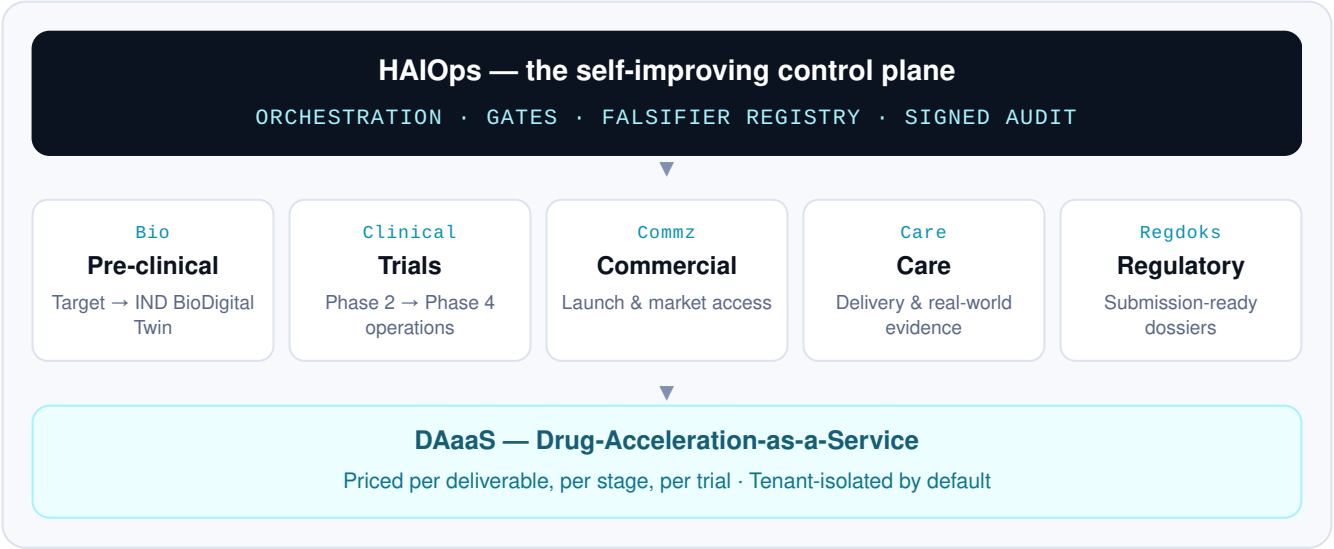
Designed around the wet-lab loop — IVIVE, PBPK and credibility frameworks — so models speak to instruments and regulators, not just dashboards.

> **THE CORE IDEA — PREDICT, THEN CONFIRM**
>
> Every Hakase output is paired with the experiment that would disprove it. When the result arrives, the loop closes automatically and the engine updates. Prediction quality compounds over time instead of decaying.

04 Platform at a glance

One foundation, five modules, end to end

Hakase is organised as a single engine with five composable modules that span the entire molecule-to-market loop. The engine — HAIOps — runs the digital workforce and enforces the falsifier registry. The modules sit on top of it; the commercial model, DAaaS, sits underneath.



Each module is documented in detail in the sections that follow. A program can adopt a single module or run the full loop; because they share one engine, value compounds as more of the pipeline moves onto the platform.

05 Hakase Bio

Pre-clinical BioDigital Twin · Discovery → IND

Hakase Bio

PRE-CLINICAL · BIODIGITAL TWIN

A zero-lab cascade from target to organism.

Hakase Bio fuses cellular simulation, structural prediction, and whole-body pharmacokinetics into a single BioDigital Twin. Hypotheses are screened, ranked, and paired with a falsifier before a single well is plated — so wet-lab time is spent confirming the most promising candidates, not exploring blindly.

MARKET — BIOTECH & PHARMA R&D · DISCOVERY → IND

What it does

Cellular AI simulation

Single-cell foundation models with integrated reference atlases, mechanistic perturbation cascades, and pathway-aware priors to predict how cells respond.

Structural × tissue gate

Predicted protein–ligand structures are gated by tissue-level expression and proteomics — candidates only progress where the underlying biology supports them.

Organism PBPK

Whole-body PK/PD with in-vitro-to-in-vivo (IVIVE) translation, first-in-human dose priors, and species bridging.

Predict-then-confirm

Every prediction ships with a falsifier and a confirmation slot. Wet-lab readouts close the loop and refine the twin automatically.

Safety & ADMET profiling

Built in to the cascade is a battery of in-silico safety and developability checks — structural alerts for mutagenicity (Ames) and screens for liver (DILI) and cardiac (hERG) liabilities, alongside a machine-learning ADMET engine covering dozens of absorption, distribution, metabolism, excretion and toxicity endpoints. Each candidate is profiled for physicochemical properties and developability before it advances, giving teams an early read on IND readiness.

How the cascade is layered

L1	**Cellular**	Single-cell foundation models and pathway-aware perturbation simulation predict cellular response to a candidate.
L2	**Structural**	Structure prediction, binder design and retrieval, gated by tissue-level expression so only biologically supported hits advance.
L3	**Organism**	PBPK and IVIVE translation engines for first-in-human dosing and cross-species bridging.

10×	**–65%**	**92%**
Hypotheses screened	**Wet-lab cycles to lead**	**Cross-assay accuracy**
Per scientist, per week	Fewer dead-end experiments	vs. held-out CROs

06 Hakase Clinical

Phase 2 → Phase 4 trial operations

Hakase Clinical — PHASE 2 → PHASE 4 OPERATIONS

Trial operations on a falsifier-first control plane.

Hakase Clinical runs protocol intelligence, site and cohort optimisation, and continuous safety-signal detection over hundreds of thousands of trial records and tens of millions of adverse-event reports. Every output is gated, auditable, and tied to a confirmation.

MARKET — CLINICAL TRIAL SPONSORS · CROS

What it does

Protocol intelligence

Compare 400,000+ public trial protocols by endpoint, dose, eligibility and adaptive design — with completion-rate benchmarks.

Monte-Carlo enrollment

Stochastic simulation delivering P10 / P50 / P90 enrollment timelines drawn from real completed-trial data.

Site & investigator intelligence

Global site and investigator ranking by enrollment velocity, start-up history and therapeutic-area experience.

Safety-signal detection

Pharmacovigilance-grade PRR / ROR detection from 20M+ adverse-event reports using established WHO methodology.

Evidence library

Full-text search across 36M+ articles with trial classification, author-level KOL ranking and abstract summaries.

Outcome prediction

A calibrated machine-learning ensemble trained on the public trials corpus delivers GO / NO-GO recommendations with confidence.

40%	$2.1M	400K+
Faster trial cycles	**Annual cost savings**	**Trial protocols**
Median across sponsors	Per Phase 2/3 program	Indexed and gated

07 Hakase Commz

Commercial intelligence for launch & market access

Hakase Commz

COMMERCIAL INTELLIGENCE

Launch-to-uptake intelligence, refreshed weekly.

Account-level scoring across payers, integrated delivery networks and KOL networks. Behavioral models for prescriber adoption and access friction, with next-best-action playbooks per territory.

MARKET — LAUNCH & MARKET-ACCESS TEAMS

What it does

Launch readiness

Account-level scoring across payers, IDNs, GPOs and KOL networks — refreshed weekly, not yearly.

HCP & payer modeling

Behavioral models for prescriber adoption and access friction, calibrated against open claims data.

Field-force orchestration

Next-best-action playbooks per territory, with explanations rep teams will actually read.

Evidence activation

Auto-generated payer dossiers and KOL talking points from real-world data — every claim traced to source.

3×	+18%	60d
Faster account prioritization	**Year-1 uptake**	**Data lake to field-ready**
	vs. consensus forecast	

08 Hakase Care

Care delivery & real-world evidence

Hakase Care

CARE DELIVERY PLATFORM

From bedside care to evidence packs, without a translation tax.

Hakase Care closes the loop between care delivery and the evidence regulators and payers need. Quality, outcomes, pathway analytics and federated real-world evidence — tenant-isolated by default.

MARKET — HOSPITALS · HEALTH SYSTEMS · PAYERS

What it does

Quality & outcomes

Disease-area dashboards aligned to CMS, HEDIS and NHS QOF frameworks.

Cohort & pathway analytics

Care-pathway mining over public and tenant data with cohort-builder semantics.

Real-world evidence

Federated cohort matching for label expansion, HEOR and post-marketing commitments.

Patient retention

Risk stratification with confirmation-tracked interventions and outcome attribution.

85%	12+	Private
Patient retention	**Disease areas**	**Tenant-isolated**
Avg. across deployments	Pre-modeled	HIPAA · GDPR by default

09 Hakase Regdoks

Regulatory submissions across four agencies

Hakase Regdoks

REGULATORY SUBMISSIONS

Submission-ready dossiers, composed from one source of truth.

Module-aware CTD/eCTD assembly across four agencies. Credibility evidence packs are auto-composed, with cross-link traceability between every model, dataset and conclusion.

MARKET — REGULATORY AFFAIRS · GCC · FDA · EMA · PMDA

What it does

Dossier assembly

Module-aware CTD/eCTD assembly with cross-link traceability between every model, dataset and conclusion.

Credibility evidence

V&V40 and MID3 evidence packs auto-generated for model-informed submissions — context of use, risk, and verification plan.

Multi-agency mapping

A single source of truth mapped to GCC, FDA, EMA and PMDA expectations, with diff views per agency.

Signed audit trail

Every prediction-to-confirmation pairing is signed, versioned and exportable for regulator review.

9mo → 9wk	100%	4
Dossier assembly cycle	**Predictions tied to a falsifier**	**Agencies pre-mapped** GCC · FDA · EMA · PMDA

10 HAIOps — the control plane

The engine that keeps the platform honest

HAIOps is the self-improving control plane shared by every module. It runs the digital workforce, enforces the falsifier registry, and stitches every prediction to its eventual confirmation. Audit trails are signed, versioned and exportable — ready for V&V40 and MID3 review.

Because orchestration, gates, governance and signed audit are promoted to a single shared plane rather than rebuilt per tool, every module inherits the same discipline. A prediction made anywhere on the platform is held to the same evidentiary standard.

- A coordinated **digital workforce** of specialist agents, swarms and BioDigital Twin curators.
- An **evidence ladder** with required confirmation slots at every step.
- **Predict-then-confirm enforcement** applied uniformly across all layers and modules.
- Signed, versioned, exportable **audit trails** built for regulatory review.

11 DAaaS — how you engage Hakase

Drug acceleration on a per-deliverable contract

Drug-Acceleration-as-a-Service prices Hakase the way teams actually consume it — per deliverable, per stage, per trial. There are no seat licenses and no shelfware. You pay for outcomes the platform produces, aligned to the milestones your program already tracks.

Aligned to outcomes

Pricing maps to deliverables and confirmations, not platform real estate. Value scales with the work, not the headcount.

Your data stays yours

Tenant-isolated by default. The engine never trains on customer data; confirmations are used as ground truth, not as training fuel.

Bench-grounded

Wet-lab, CRO and animal readouts are treated as ground truth that closes the loop and improves predictions over time.

Agency-ready

Outputs are mapped to GCC, FDA, EMA and PMDA expectations, so evidence is submission-ready without a translation tax.

12 Security, privacy & compliance

Privacy-first defaults, built for regulated environments

Hakase is designed for the data-sensitivity and governance demands of pharma, clinical and healthcare settings. Isolation, auditability and provenance are defaults rather than add-ons.

Tenant isolation by default

Each customer's environment is isolated. Data is not pooled across tenants, and the engine never trains on customer data.

Confirmations, not training fuel

Experimental and clinical readouts are used strictly as ground truth to close loops — never absorbed into shared model weights.

Signed, exportable audit

Every prediction-to-confirmation pairing is signed and versioned, producing a defensible trail for internal and regulatory review.

Regulatory frameworks

Built to align with established credibility and quality frameworks for model-informed evidence and care-quality reporting.

HIPAA · GDPR · GxP · V&V40 credibility · MID3 evidence · CTD / eCTD · GCC · FDA · EMA · PMDA mapped

Frameworks listed describe the standards Hakase is designed to support. Specific certifications and attestations for a given deployment are confirmed during onboarding.

13 Outcomes & who it's for

The numbers and the teams that benefit

Headline outcomes

10×	40%	$2.1M	9mo→9wk
Hypotheses screened	**Faster trial cycles**	**Annual savings**	**Dossier assembly**
Per scientist, per week (Bio)	Median across sponsors	Per Phase 2/3 program	Regulatory cycle

Built for every stage of the pipeline

Biotech & pharma R&D

Discovery and pre-clinical teams screening and de-risking candidates toward IND.

Sponsors & CROs

Clinical operations teams designing protocols, choosing sites and watching safety signals.

Launch & market access

Commercial teams prioritising accounts and activating evidence for payers.

Hospitals & health systems

Care teams turning delivery into quality, outcomes and real-world evidence.

Regulatory affairs

Teams assembling submission-ready dossiers across four agencies.

Payers

Organisations seeking traceable, evidence-backed outcomes and access decisions.

14 Get in touch

A working session, not a sales call

The best way to evaluate Hakase is to see a real falsifier run on a problem you care about. Tell us what you are working on and we will set up a working session walking through a live prediction-then-confirm loop on representative data.

BRING HAKASE TO YOUR NEXT PROGRAM

One engine. Five modules. Every prediction, confirmable.

Whether you are de-risking a discovery candidate, designing a trial, planning a launch, or assembling a dossier — Hakase composes the same trustworthy engine for the job.

REQUEST A DEMO

A guided walkthrough on your use case

PARTNER WITH US

Co-develop on a live program

WEB

HakaseAI · the operating system for drug development

This document is intended for client and partner distribution. It describes platform capabilities and indicative outcomes; specific results, scope and commercial terms are defined per engagement. Figures cited are indicative platform benchmarks, not a guarantee of results for any particular program.